

Effect of Continuous Transition between Force Stimuli on Haptic Force Discrimination Tasks

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Introduction to Force Discrimination

- Haptic perception enables humans to sense and interpret the physical world through touch. This work focuses on **kinesthetic force perception**.
- Perceptual limitations for force stimuli are quantified by the **Just Noticeable Difference (JND)** and the **Weber fraction (δ)**¹.
- Conventional force discrimination tasks involve comparing pairs of forces separated by an inter-stimulus interval (ISI).

- **Weber Fraction:**

$$\delta = \frac{|F_t - F_r|}{|F_r|} \quad (1)$$

- F_r : reference force • F_t : test force

- **Perceptual Deadband:**

$$[(1 - \delta)F_r, (1 + \delta)F_r] \quad (2)$$

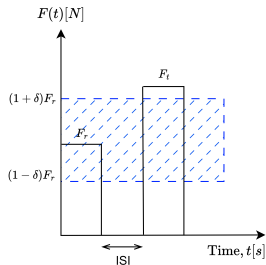


Fig. 1. Conventional Force Discrimination with ISI

¹L. A. Jones et al., 2012

Objective

- It is unclear whether the force JND/Weber fraction remains consistent for **Continuous Force Perception** when measured with conventional methods (with ISI).

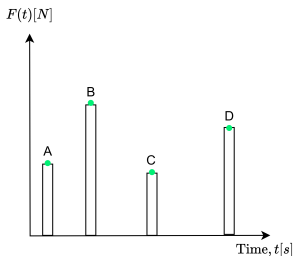


Fig. 2(a). Discrete Force

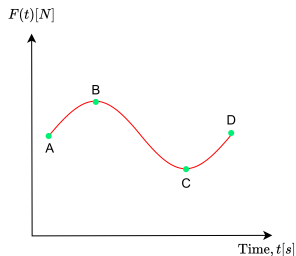


Fig. 2(b). Continuous Force

- In this study:** We determine how the continuous transition between force stimuli affects force perception.

A. Experimental Setup

- Device: 'PHANToM Premium 1.5 HF'
- Force stimulus $\rightarrow +Z$ -axis



Fig. 3. Experimental setup

B. Force Stimuli

- Two transition conditions:
 - a Time-separated transition
 - b Continuous linear transition

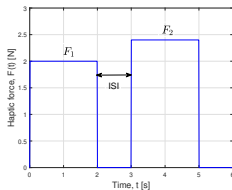


Fig. 4(a). Time-separated

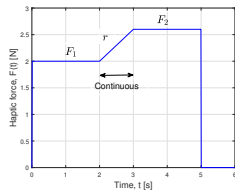


Fig. 4(b). Continuous linear

Methodology (cont.)

- One of F_1 or F_2 is set as reference force and the other as comparison force.
- Reference force $F_r = 2\text{N}$ and Comparison force F_c :
 - a Time-separated: $F_{c1} = \{1.2, \dots, 2.8\}\text{N}$ (9 levels)
 - b Continuous: $F_{c2} = \{1.0, \dots, 3.0\}\text{N}$ (11 levels)

C. Procedure

- Classical Psychophysical Method: Method of Constant Stimuli
- Paradigm: 2-Alternative Forced Choice (2-AFC)
- Participants: Same 10 users (8M + 2F)
- Total trials: More than 1000 trials per user

Results

Psychometric Functions: by fitting Cumulative Gaussian Functions²

$$\bullet \text{ JND} = P_{84} - P_{50}, \delta = \frac{\text{JND}}{F_r}$$

$$\bullet \text{ PSE} = P_{50}$$

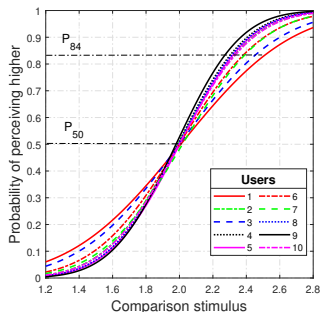


Fig. 5(a). Psychometric fit – Time-separated condition

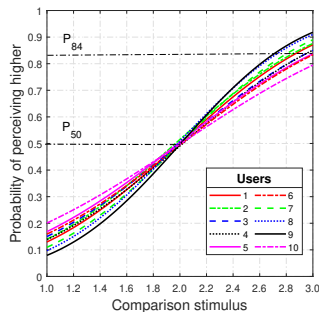


Fig. 5(b). Psychometric fit – Continuous condition

²K. Zchaluk et al., 2009

Results (cont.)

Weber fraction range (across all users):

- Time-separated: [14.97, 25.87]%
- Continuous linear: [35.44, 50.55]%

Statistical Results: Paired sample t-test results:

- WF: $t(9) = -14.80, p = 0.000$
- PSE: $t(9) = -0.23, p = 0.824$

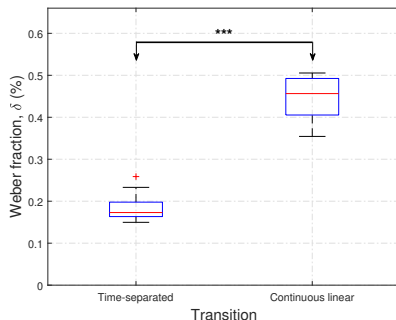


Fig. 6. Comparison of Weber fraction across conditions

- Equal PSE $\rightarrow$ both user accuracy and experimental validity, showing reliable and unbiased performance for both conditions.
- Weber fraction for the time-separated transition aligns closely with established findings in force perception^{3 4}
- Mean WF: continuous (44.63%) $>$ time-separated (18.64%)
- Suggests users are more sensitive to continuous transitions

³X.-D. Pang et al., 1991

⁴S. Allin et al., 2002

Application

- Continuous Force Perception in haptic data compression for teleoperation.
- This allows more data compression since the Weber fraction is higher than the time-separated condition.
- However, to reflect continuous force perception, gradual transitions are required (e.g., linear interpolation—blue dashed in Fig. 7b).

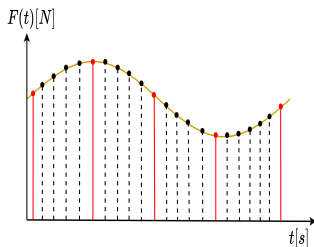


Fig. 7(a). Transmitter side (brown color)

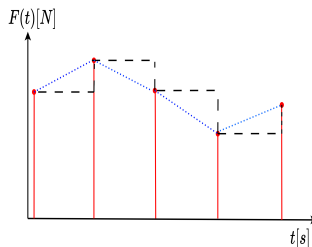


Fig. 7(b). Receiver side

Conclusion and Future Work

Conclusion

- Continuous force transitions result in a significantly higher Weber fraction compared to time-separated conditions.
- These findings indicate that force discrimination becomes more difficult when transitions are continuous.
- Implications for haptic system design: Continuous force perception must be considered when optimizing for realism or compression.

Future Work

- Compare continuous linear transition (current study) with a case where ISI is 0.
- Effect of different reference forces (F_r) on continuous force JND/WF.
- Explore the effect of active movement in addition to the current passive setup.

Thank You!

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